Project Title: Neurocognitive Responses to Exam Questions: A Pilot Study Using fNIRS and BCI

Investigators

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What is this study about?

We want to understand how students' brains work when answering different types of exam questions. We will use a safe, non-invasive brain measurement tool called **functional near-infrared spectroscopy (fNIRS)**. The results may help improve how exams and learning tools are designed. This study is for research only and will not give you any medical results.

How does fNIRS work? It uses harmless near-infrared light to measure tiny changes in blood flow in your brain while you think. This helps researchers see which areas of your brain are more active when you answer different types of questions. The device does not emit any harmful radiation, does not affect your brain, and is completely painless.

What will I need to do?

If you agree to take part:

- You will take part in a single session lasting about **1 hour**.
- You will wear a soft, light cap that measures your brain activity — this does not hurt and is completely safe.
- You will answer a set of exam-style questions on a computer.
- You will fill in a short form about yourself (like age, study level, and your parents' occupation).

There are no right or wrong answers. You can skip any question if you feel uncomfortable.

Are there any risks or benefits?

There are no known risks. The fNIRS device does not use radiation and does not affect your health. While you may not benefit personally, your participation will help improve education research.

Do I get anything for taking part?

Yes. You will receive a one-time payment of **€50** to thank you for your time.

Will my information be kept private?

- All the information you give will be kept **private and confidential**.
- Your name will not appear in any reports or publications.
- Your data will be given a code number and stored securely.
- Only the research team will have access to your coded data.
- In line with GDPR, you can ask to see your data, ask us to correct it, or ask for it to be deleted at any time.
- Your data will be kept for up to **5 years** for research and possible audits, then permanently deleted.

Important:

This study does not give medical results. If you want information about your health, please see your doctor. We will not be able to interpret any unexpected findings.

Is participation voluntary?

Yes. Taking part is completely your choice. You can say no or stop at any time without giving a reason. If you stop, your data will not be used.

Your Basic Rights under the EU General Data Protection Regulation (GDPR)

Voluntariness and Withdrawal (Art. 7(3) GDPR): Providing your personal data is voluntary. You can withdraw your consent for the processing of your personal data at any time. If you do, any further processing of your data will stop.

Right of Access (Art. 15 GDPR): You have the right to request information at any time about which of your personal data is being processed and who may receive this data. You will receive a response within one month of your request.

Right to Rectification, Erasure, and Restriction (Art. 16–18 GDPR): You can ask LISER to correct, delete, or restrict the processing of your personal data at any time.

Right to Data Portability (Art. 20 GDPR): You may request that your personal data be provided to you in a machine-readable format or transferred directly to another responsible party, where technically possible.

Right to Lodge a Complaint (Art. 77 GDPR): You can contact LISER's Data Protection Officer at any time. If you wish to file a complaint under Art. 77 GDPR, you can also contact the competent data protection supervisory authority.

How to exercise your rights: To exercise any of these rights, simply send a written notification (letter or email) to the responsible person or the Data Protection Officer.

Who can I contact?

If you have questions, please contact: **Thiago Brant (Principal Investigator)**:
`thiago.Quaresmabrantchaves@liser.lu`
For questions about how your data is protected: **LISER Data Protection Officer**:
`dpo@liser.lu`

Consent Form

Please read each statement and tick the box to show that you agree.

- ☐ I am **18 years old or older**.
- ☐ I have read and understood the information about the study.
- ☐ I have had the chance to ask questions and received clear answers.
- ☐ I understand what I will need to do if I take part.
- ☐ I know I can withdraw at any time without any problem.
- ☐ I know my data will be kept confidential and securely stored.
- ☐ I agree to take part in this research.

Participant Name:	**Date:**
Participant Signature:	**Researcher Signature:**

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